

The Role of Equatorial Rossby Waves in Tropical Cyclogenesis. Part I: Idealized Numerical Simulations in an Initially Quiescent Background Environment

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ABSTRACT

This two-part series of papers examines the role of equatorial Rossby (ER) waves in tropical cyclone (TC) genesis. To do this, a unique initialization procedure is utilized to insert $n = 1$ ER waves into a numerical model that is able to faithfully produce TCs. In this first paper, experiments are carried out under the idealized condition of an initially quiescent background environment. Experiments are performed with varying initial wave amplitudes and with and without diabatic effects. This is done to both investigate how the properties of the simulated ER waves compare to the properties of observed ER waves and explore the role of the initial perturbation strength of the ER wave on genesis. In the dry, frictionless ER wave simulation the phase speed is slightly slower than the phase speed predicted from linear theory. Large-scale ascent develops in the region of low-level poleward flow, which is in good agreement with the theoretical structure of an $n = 1$ ER wave. The structures and phase speeds of the simulated full-physics ER waves are in good agreement with recent observational studies of ER waves that utilize wavenumber–frequency filtering techniques. Convection occurs primarily in the eastern half of the cyclonic gyre, as do the most favorable conditions for TC genesis. This region features sufficient midlevel moisture, anomalously strong low-level cyclonic vorticity, enhanced convection, and minimal vertical shear. Tropical cyclogenesis occurs only in the largest initial-amplitude ER wave simulation. The formation of the initial tropical disturbance that ultimately develops into a tropical cyclone is shown to be sensitive to the nonlinear horizontal momentum advection terms. When the largest initial-amplitude simulation is rerun with the nonlinear horizontal momentum advection terms turned off, tropical cyclogenesis does not occur, but the convectively coupled ER wave retains the properties of the ER wave observed in the smaller initial-amplitude simulations. It is shown that this isolated wave-only genesis process only occurs for strong ER waves in which the nonlinear advection is large. Part II will look at the more realistic case of ER wave–related genesis in which a sufficiently intense ER wave interacts with favorable large-scale flow features.

1. Introduction

a. Overview

Diabatic heating within regions of tropical convection may excite various zonally propagating equatorial wave motions. Understanding the role of these equatorially trapped, tropical waves is fundamental to understanding tropical dynamics, and ultimately tropical cyclone (TC) genesis. Since 80%–90% of all tropical cyclones form

within 20° of the equator (Frank and Roundy 2006), equatorially trapped, tropical waves may ultimately influence TC genesis over much of the globe.

The purpose of this study is to examine the horizontal and vertical structure of a meridional mode number 1 ($n = 1$) equatorial Rossby (ER) wave as well as the wave's TC genesis potential using a series of model initial-value experiments with a resolution capable of efficiently modeling both the large-scale features of a propagating ER wave and the process of TC genesis. This study is designed to address a few key questions. First, are the dry, frictionless simulation results similar to what is expected from shallow-water theory? Second, how does the structure and phase speed of the $n = 1$ ER wave

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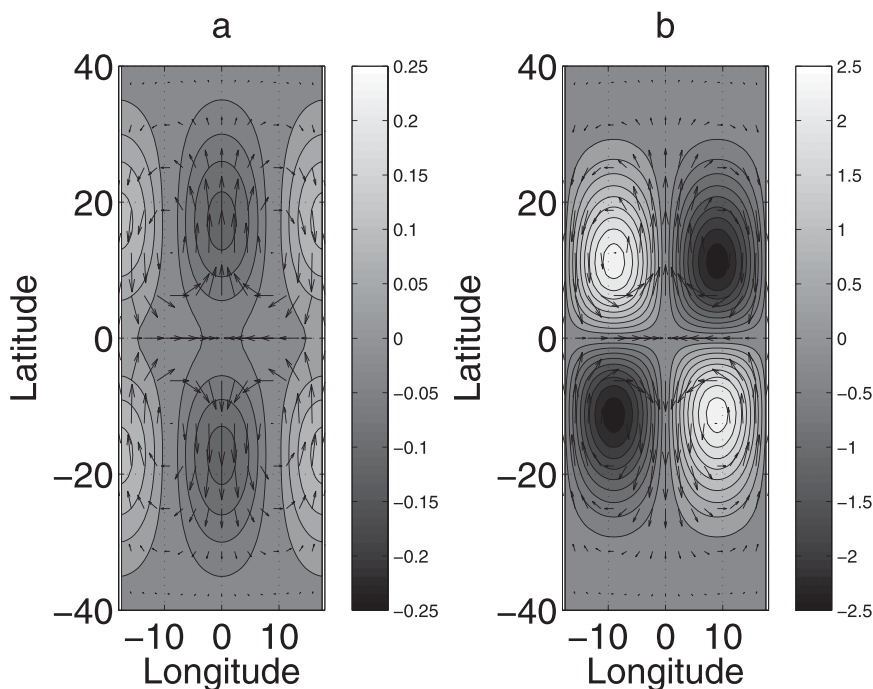


FIG. 1. The dimensionalized 850-mb wind vectors and (a) dimensional divergence (10^{-5} s^{-1}) at 850 mb with a contour interval of $0.025 \times 10^{-5} \text{ s}^{-1}$, and (b) dimensional relative vorticity (10^{-5} s^{-1}) at 850 mb with a contour interval of $0.25 \times 10^{-5} \text{ s}^{-1}$ for the $n = 1$ ER wave solution to the shallow-water equations on an equatorial β plane plotted over one wavelength of the ER wave. The magnitude of the maximum wind vector is 12.8 m s^{-1} . The solutions are based on a planetary wavenumber 10 structure, a Rossby radius (L) of 1391 km, an equivalent depth of 200 m, and an amplitude A of 0.16. For further information, refer to section 2b.

change when diabatic heating, surface fluxes, and friction are included in the simulation? Third, what are the magnitudes of the circulations associated with the ER wave (e.g., low-level convergence, vorticity, and vertical shear) and how significant are these circulations with respect to TC genesis? Finally, is an ER wave capable of resulting in TC genesis owing to its anomalous circulations alone (i.e., with no background flow interactions)? It is believed that the methodology presented herein provides a unique tool to address the previously posed questions, and has advantages over a methodology that utilizes wave-filtered observations.

b. Background

Matsuno (1966) provided the first comprehensive theoretical understanding of zonally propagating, equatorially trapped tropical waves. The theoretical dispersion relationship was derived for eastward- and westward-propagating inertial-gravity waves, westward-propagating ER waves, eastward-propagating Kelvin waves, and mixed Rossby-gravity waves. These classic equatorial waves are either symmetric or antisymmetric about the equator. In particular, the structure of the $n = 1$ ER wave is dominated by the rotational component of the wind as

demonstrated by Delagen and Yano (2009) and as seen in a comparison of the divergence (Fig. 1a) to relative vorticity (Fig. 1b). For the $n = 1$ ER wave, the magnitude of the wind and geopotential are quite strong, while the divergence is relatively weak when compared to values for other theoretical wave types (Wheeler 2002).

While the work of Matsuno (1966) laid the theoretical framework for tropical waves, various observational studies have verified that these waves exist within the tropics and are significant components of tropical weather. In one of the first observational papers on ER waves, Kiladis and Wheeler (1995) demonstrated that ER waves have a maximum anomalous signal in the lower troposphere, are associated with convective signals at roughly the mean latitude of the tropical convergence zones, and possess many of the features of the analytic $n = 1$ ER wave mode derived by Matsuno. The authors found that ER waves feature, on average, a wavenumber 6 zonal scale and a deep, nearly equivalent barotropic structure up to 100 mb.

Wheeler and Kiladis (1999) utilized a wavenumber-frequency spectral analysis of satellite-observed outgoing longwave radiation (OLR), a proxy for cloudiness, in order to separate phenomena in the time-longitude domain

into westward- and eastward-moving components. They found that several statistically significant spectral peaks in the wavenumber–frequency spectra exist, one of which was the $n = 1$ ER wave. In Wheeler et al. (2000), the large-scale dynamical fields associated with convectively coupled equatorial waves were examined. In particular, their composite ER wave had a westward phase speed of 5 m s^{-1} , a wavenumber 5 zonal scale, and enhanced convection and low-level convergence in the region equatorward and eastward of the center of the cyclonic gyre of the ER wave. The observed location of maximum low-level convergence was shifted somewhat westward and equatorward compared to the inviscid theoretical shallow-water structure. The observational study of Roundy and Frank (2004a) found that convectively coupled ER waves explain as much as 40% of the seasonal convective variance within certain regions of the tropics.

Frank and Roundy (2006) analyzed relationships between TC formation and tropical wave activity in each of the six global basins. Five wave types were examined in their study, including mixed Rossby–gravity waves, tropical depression-type or easterly waves (TD type), ER waves, Kelvin waves, and the Madden–Julian oscillation (MJO; e.g., Madden and Julian 1994; Zhang 2005). Composite analyses were constructed relative to the storm genesis locations for each of the five wave types in order to show the structure of the waves and their preferred phase relationships with the genesis location. It was found that all of the wave types except for Kelvin waves play a significant role in TC formation by creating an environment favorable for TC genesis. Their composite analysis for the ER wave filter band in the northwest Pacific showed a strong cyclonic gyre centered just northwest of the genesis location with maximum convection about one-quarter wavelength to the east of the center of the cyclonic gyre. The preferred region of genesis with respect to the ER wave was located equatorward and eastward of the ER wave gyre center in the region of anomalous cyclonic flow and negative OLR anomalies (enhanced convection).

Bessafi and Wheeler (2006) analyzed the relationships between various tropical wave types and TC genesis over the southern Indian Ocean. Analysis of all TCs west of 100°E revealed a large and statistically significant modulation by ER waves. For the ER wave, TC modulation was attributed to perturbations of the convection and vorticity fields. The magnitude of the maximum vorticity anomalies associated with the ER wave was on the order of $5 \times 10^{-6} \text{ s}^{-1}$. Bessafi and Wheeler (2006) also examined vertical shear modulations within the ER wave. They found an almost equal number of TCs forming on either side of the zero zonal shear anomaly line, and concluded that vertical shear modulation was less important

than the anomalous low-level vorticity or convection associated with the ER wave.

Molinari et al. (2007) identified a packet of ER waves that lasted 2.5 months in the lower troposphere of the northwest Pacific that appeared highly influential in a number of tropical cyclogenesis events. The ER waves within the packet had a wavelength of 3600 km (zonal wavenumber 11) and a westward zonal phase speed of 1.9 m s^{-1} . It should be noted that the zonal wavenumber of this ER wave packet was much greater than that observed in previous observational studies of ER waves. The wave properties followed the ER wave dispersion relation for an equivalent depth near 25 m. The authors found that the packet was associated with the development of at least 8 of the 13 tropical cyclones that formed during the period. Unfiltered OLR and unfiltered 850-mb wind and vorticity were composited with respect to the genesis location of the ER wave–related tropical cyclones. The mean genesis location occurred in a region of enhanced convection (negative OLR anomalies) within an area of anomalous low-level cyclonic vorticity. In this case, the mean genesis location was east, and slightly equatorward of, the ER wave gyre center. Molinari et al. (2007) also composited the unfiltered 200–850 mb vertical shear with respect to the genesis location. The mean genesis location resided in a region of weak vertical shear with a magnitude of less than 10 m s^{-1} . The authors concluded that the positive impacts of ER wave–induced convection and cyclonic vorticity were of greater importance than those of ER wave–induced vertical wind shear.

c. Outline

Section 2 of this paper features a description of the model used, the method for inserting an ER wave into the model initial condition, and outlines the five experiments performed. Section 3 presents results from the various experiments. Section 4 provides a discussion of the results, additional avenues for future work, and a motivation for Part II (Gall and Frank 2010, hereafter Part II) in which the role of a background flow is investigated for ER wave–related genesis.

2. Methodology

a. Model setup

A tropical strip model was designed using the Weather Research and Forecasting (WRF) model version 2.1.1. WRF is a next-generation, regional, fully compressible model of the atmosphere presently under development by a number of agencies involved in atmospheric research and forecasting (Michalakes et al. 2001). The model domain has a grid spacing of 81 km with 493×117

grid points in the horizontal and 31 unevenly spaced vertical levels. Such a configuration results in a domain that extends around the entire globe between 38°N and 38°S latitude with periodic boundary conditions in the x direction and rigid walls at the north and south boundaries. It is argued that the boundary conditions at the north and south borders are sufficient since the meridional wind component of equatorially trapped waves decays toward zero away from the equator. All terrain was removed, and the entire surface skin mask ($z = 0$) was set to water such that the model was run as an aquaplanet. The large time step used was 200 s, which ensured numerical stability. The model domain featured variable Coriolis parameter and a constant sea surface temperature (SST) set to 28.5°C.

A six-species cloud microphysics package was used, which included water vapor, rainwater, cloud water, cloud ice, snow, and hail–graupel (Lin et al. 1983). The modified version of the Kain–Fritsch scheme (KF-Eta) was used to parameterize convective processes. This scheme is based on Kain and Fritsch (1990, 1993), but has been modified based on testing within the Eta model. As with the original KF scheme, it utilizes a simple cloud model with moist updrafts and downdrafts, including the effects of detrainment, entrainment, and relatively crude microphysics (Chen and Dudhia 2000). The atmospheric boundary layer was parameterized using the Yonsei University (YSU) scheme. This scheme is similar to the Medium-Range Forecast (MRF) scheme (Hong and Pan 1996) in that it uses a so-called countergradient flux for heat and moisture in unstable conditions, enhanced vertical flux coefficients in the boundary layer, and handles vertical diffusion with an implicit local scheme. The scheme also explicitly treats entrainment processes at the top of the entrainment layer (Hong and Pan 1996; Hong et al. 2004; Noh et al. 2004). The Monin–Obukhov surface layer scheme was used to compute the surface exchange coefficients for heat, moisture, and momentum.

In this study, no specific radiation scheme available in the WRFv2.1.1 package was employed. Rather, a constant radiational cooling of -0.5 K day^{-1} was applied at all vertical model levels. This was done because the available radiation parameterizations were designed for real-data simulations. Given the idealized configuration, the model domain is not in energetic or moisture balance with the true radiational cooling expected in the real atmosphere, and the use of an interactive radiation parameterization will cause the domain to drift from realistic tropical conditions. The choice of the constant value of -0.5 K day^{-1} was used since this cooling rate produces a relatively steady, domain-averaged temperature and moisture profile for the numerical simulations conducted in this study. Furthermore, the -0.5 K day^{-1}

radiational cooling rate is a good approximation to observed radiational cooling rates within the tropics (e.g., Holton 2004). One limitation of the fixed radiational cooling, however, is that the scheme effectively suppresses longwave cloud-radiative feedbacks. As demonstrated by Bretherton et al. (2005), this mechanism plays a significant role in the self-aggregation of deep, moist convection. Since the aggregation of convective-scale vorticity is a potential pathway to tropical cyclogenesis (e.g., Reasor et al. 2005), it is possible that the specified radiation scheme may inhibit tropical cyclogenesis. The sensitivity of the structure of the simulated ER waves to the radiation scheme is beyond the scope of the current study and will be examined in a future study.

b. ER wave initialization

This section provides the derivation of the three-dimensional structure of an $n = 1$ ER wave used in the initial condition of the WRF model. We develop this initial condition from linear shallow-water theory. As will be shown, this theoretical structure serves as a useful means to insert an $n = 1$ ER wave into the model initial condition despite the simplifications involved in the theory compared to the model. First, the horizontal, non-dimensional solutions for an $n = 1$ ER wave are provided. Then, the procedure for dimensionalizing the horizontal solutions is discussed, and finally, the method for specifying the vertical variation of the initial ER wave structure is presented.

Following the work of Matsuno (1966), the set of shallow-water equations can be made nondimensional through use of a length scale (L):

$$L = \sqrt{\frac{c}{\beta}} \quad (1)$$

and time scale (T):

$$T = \frac{1}{\sqrt{c\beta}}, \quad (2)$$

where c is the gravity wave speed and β is the planetary vorticity gradient. Here c is given by

$$c = \sqrt{gh}, \quad (3)$$

where g is gravity and h is the equivalent depth. It can then be shown that the nondimensional, shallow-water, meridional wind, geopotential, and zonal wind perturbations for an ER wave of nondimensional wavenumber k^* may be given by

$$v^*(x^*, y^*) = AH(n, y^*) \exp\left(-\frac{y^{*2}}{2}\right) \cos(k^*x^*), \quad (4)$$

$$\begin{aligned} \phi^*(x^*, y^*) = & A \frac{-\exp\left(-\frac{y^{*2}}{2}\right)}{(\omega^{*2} - k^{*2})} \\ & \times [k^* y^* H(n, y^*) - 2n\omega^* H(n-1, y^*) \\ & + \omega^* y^* H(n, y^*)] \times \sin(k^* x^*), \end{aligned} \quad (5)$$

$$\begin{aligned} u^*(x^*, y^*) = & A\omega^{*-1} \left[k^* \phi^*(x^*, y^*) \right. \\ & \left. - \frac{y^* v^*(x^*, y^*) \sin(k^* x^*)}{\cos(k^* x^*)} \right], \end{aligned} \quad (6)$$

where the * indicates nondimensionality, and x and y are distances in the eastward and northward directions, respectively, A controls the amplitude of the perturbation,¹ ω is the frequency, ϕ is the geopotential perturbation, and $H(n, y^*)$ is the nondimensional Hermite polynomial. The first two Hermite polynomials are given by

$$\begin{aligned} H(0, y^*) &= 1 \\ H(1, y^*) &= 2y^*. \end{aligned} \quad (7)$$

Here k^* is calculated using

$$k^* = \frac{aL}{R_e}, \quad (8)$$

where a is the planetary zonal wavenumber and R_e is the radius of the earth. The ω^* for ER waves is given by

$$\omega^* = -\frac{k^*}{k^{*2} + (2n+1)}. \quad (9)$$

Using the values provided in Table 1 gives $k^* = 2.18$ and $\omega^* = -0.28$, with $\omega^* < 0$ indicating westward propagation.

The nondimensional lengths x^* and y^* are dimensionalized using

$$x = Lx^* \quad (10)$$

and

$$y = Ly^*, \quad (11)$$

where L is on the order of 12.5° latitude for the parameters provided in Table 1. The dimensionalized expressions for v , ϕ , and u for the wavenumber 10 ER wave are given by multiplying Eqs. (4), (5), and (6) by the necessary form of c :

$$v(x, y) = cv^*(x^*, y^*) = AcH(n, y^*) \exp\left(-\frac{y^{*2}}{2}\right) \cos(k^* x^*), \quad (12)$$

$$\begin{aligned} \phi(x, y) = & c^2 \phi^*(x^*, y^*) = Ac^2 \frac{-\exp\left(-\frac{y^{*2}}{2}\right)}{\omega^{*2} - k^{*2}} \\ & \times [k^* y^* H(n, y^*) - 2n\omega^* H(n-1, y^*) \\ & + \omega^* y^* H(n, y^*)] \times \sin(k^* x^*), \end{aligned} \quad (13)$$

$$\begin{aligned} u(x, y) = & cu^*(x^*, y^*) \\ = & Aco\omega^{*-1} \left[k^* \phi^*(x^*, y^*) - \frac{y^* v^*(x^*, y^*) \sin(k^* x^*)}{\cos(k^* x^*)} \right]. \end{aligned} \quad (14)$$

Figures 2a–c shows the dimensional forms of v , ϕ , and u for a planetary zonal wavenumber 10 ER wave on the dimensionalized x – y domain. A wavenumber 10 structure was specified because this value falls within a planetary zonal wavenumber range based on Molinari et al. (2007) ($a = 11$) and Kiladis and Wheeler (1995) ($a = 6$).

The vertical structure for v , ϕ , and u were given by multiplying the solution obtained from Eqs. (12)–(14) by the particular internal mode's vertical structure function $G(z)$. That is,

$$v(x, y, z) = v(x, y)G(z), \quad (15)$$

$$\phi(x, y, z) = \phi(x, y)G(z), \quad (16)$$

$$u(x, y, z) = u(x, y)G(z). \quad (17)$$

Following the derivation of Wheeler (2002), $G(z)$ is given by

$$G(z) = \exp\left(\frac{z}{2H_s}\right) \exp(-imz), \quad (18)$$

where H_s is the scale height and m is the vertical wavenumber defined as

$$m = \frac{2\pi}{L_z} = \left(\frac{N^2}{gh} - \frac{1}{4H_s^2} \right)^{1/2}, \quad (19)$$

where L_z is the vertical wavelength of the normal mode. Here N^2 is given by

$$N^2 = \frac{R}{H_s} \left(\frac{d\bar{T}}{dz} + \frac{g}{c_p} \right), \quad (20)$$

where $d\bar{T}/dz$ is an average lapse rate, R is the gas constant, and c_p is the specific heat for dry air. Equation (19)

¹ Since these are linear solutions, we may multiply the solution by a scaling factor.

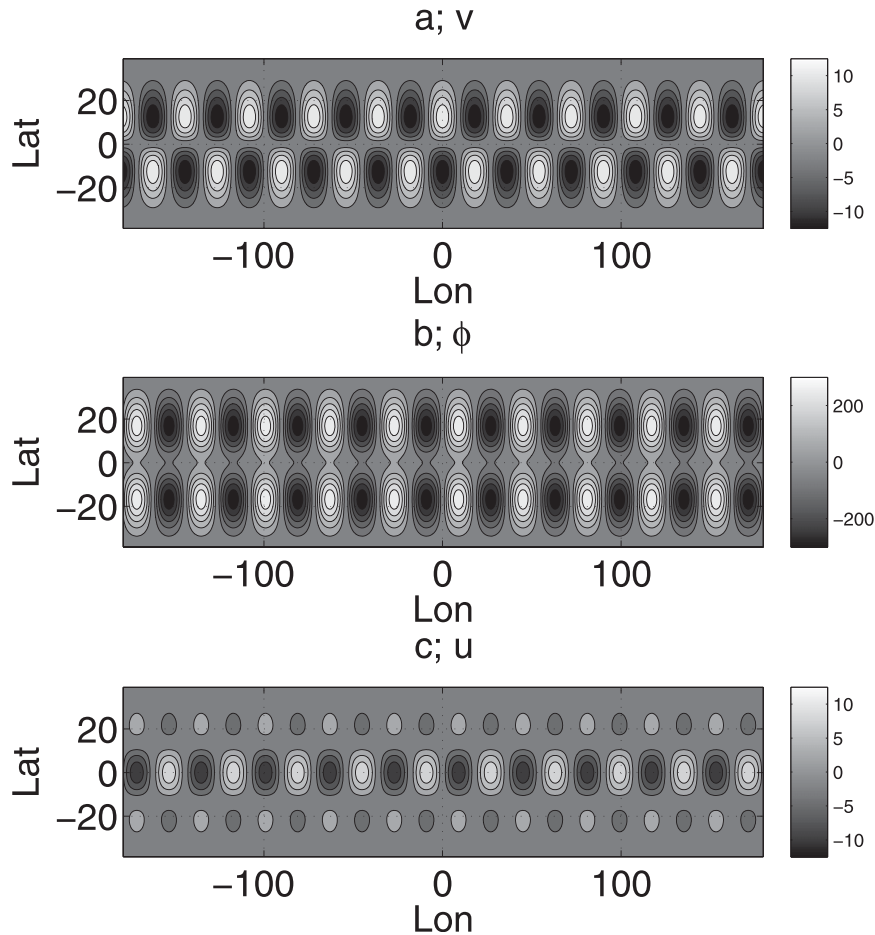


FIG. 2. Dimensional (a) v (m s^{-1}), (b) ϕ ($\text{m}^2 \text{s}^{-2}$), and (c) u (m s^{-1}) given by Eqs. (12)–(14) and the values in Table 1 for a planetary zonal wavenumber 10, $n = 1$ ER wave with $A = 0.16$. Both (a) and (c) have a contour interval of 2.5 m s^{-1} while (b) has a contour interval of $50 \text{ m}^2 \text{s}^{-2}$.

provides a relationship between the vertical wavelength of a normal mode in a constant N atmosphere, and its equivalent depth h . Even though the numerical model is not constrained to have a constant N atmosphere, providing an initial ER wave with a vertical structure specified by these theoretical relations is sufficient. The specific (baroclinic) vertical structure is shown in Fig. 3, based on the parameters provided in Table 1.

The wind field for the initial condition was generated by adding the u and v perturbations associated with the ER wave [Eqs. (15) and (17)] to the base-state wind field. Since the initial base-state winds are zero, the entire u and v structure is given by the ER wave perturbation winds. The Jordan (1958) mean hurricane season soundings of moisture and temperature were used to provide a base-state moisture and temperature profile. Through vertical integration of the hydrostatic equation, and use of these soundings, a hydrostatic base-state

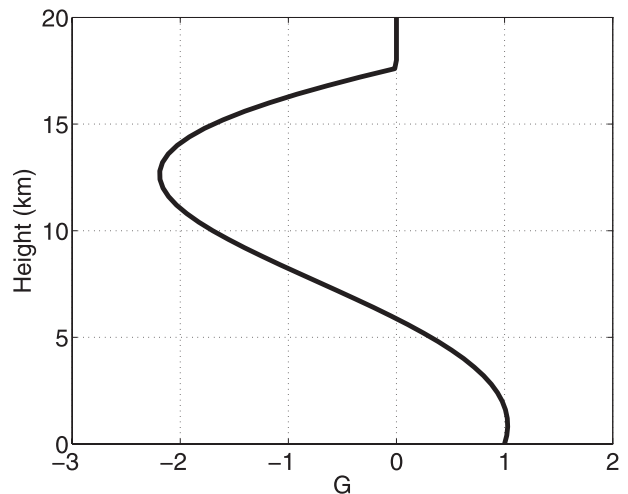


FIG. 3. Variation of G with height. G has been set to 0 above 18 km.

TABLE 1. Summary of the relevant parameters for the nondimensional and dimensional ER wave structure equations. Both H_s and $d\bar{T}/dz$ were calculated using the Jordan (1958) temperature sounding and the values for certain parameters provided in this Table.

n	a	h (m)	c (m s^{-1})	β (s^{-1})	L (km)	k^*	ω^*	H_s (km)	$d\bar{T}/dz$ (K km^{-1})
1	10	200	44.3	2.3×10^{-11}	1390	2.18	-0.28	7.7	-6.0

pressure profile was calculated. The ER wave geopotential anomalies were converted to pressure perturbations via the hydrostatic approximation and then added to the base-state pressure field. Although the initial condition was not in “model balance,” it represents a good first guess for such a balance as evidenced by the lack of gravity wave noise present in the simulations.

c. Experimental design

Five ER wave simulations are run in total, as summarized in Table 2. In simulations ER-1, ER-2, and ER-3, the initial amplitude of the ER wave is controlled via the parameter A . Here A is set to 0.09 in ER-1, 0.16 in ER-2, and 0.23 in ER-3. Since the ER wave solutions in Eqs. (15)–(17) are multiplied by A , this parameter is a means by which the initial amplitude of the ER wave is controlled. The effect of the parameter A on the initial structure of the 850-mb zonal wind and 850-mb relative vorticity is illustrated in Figs. 4a,b. The initial ER-3 meridional structure is considered to be an upper bound on ER wave intensity as ER waves with a larger initial amplitude would satisfy the necessary condition for barotropic instability. The resulting maximum relative vorticity values of $1.75 \times 10^{-5} \text{ s}^{-1}$ and $1.0 \times 10^{-5} \text{ s}^{-1}$ for ER-2 and ER-3, respectively, are based on a blend of the magnitude of the Molinari et al. (2007) ER band-filtered anomalies and the fact that composite analyses underestimate anomaly magnitudes when compared with the anomaly magnitudes from individual cases. ER-D-2 is the same as ER-2 except that this simulation features a “dry” initial condition (i.e., the initial moisture fields were set to zero) and all diabatic effects (surface fluxes, radiation, phase changes, and friction) were turned off.² ER-3-NOADV is the same as ER-3 except that the horizontal momentum advection terms are set to zero, that is, $\mathbf{v}_H \cdot \nabla \mathbf{v} = 0$. All five simulations are integrated forward in time for 30 days.

3. Results

a. Dry ER wave simulation (ER-D-2)

Figure 5a shows the 30-day Hovmöller diagram of the 850-mb meridional wind for ER-D-2. A well-defined, westward-propagating signal is evident in the v component of the wind despite that no wavenumber–frequency filter has been used in the construction of the Hovmöller diagram. In the ER-D-2 simulation, the zonal wavenumber 10 ER wave structure remains intact over the entire 30-day simulation, and propagates to the west with a speed of 3.5 m s^{-1} . The westward propagation of the ER wave in ER-D-2 is not surprising, as linear theory predicts such a result for a zero background flow environment.

The baroclinic vertical structure of the dry ER wave is maintained throughout the simulation, as seen in the plots of the 850- and 200-mb wind field at $t = 30$ days (Figs. 6a,b, respectively). That is, regions of 850-mb cyclonic (anticyclonic) flow are located beneath regions of 200-mb anticyclonic (cyclonic) flow. ER-D-2 also provides an explanation of the large-scale vertical velocity patterns. As seen in Figs. 6a,b, the region of maximum ascent (subsidence) lags the 850-mb cyclonic (anticyclonic) gyre by about a quarter wavelength. That is, the maximum large-scale ascent (subsidence) occurs within the region of low-level poleward (equatorward) flow. Shallow-water theory predicts the maximum low-level convergence, and by mass continuity, maximum ascent east of the low-level cyclonic gyre, as seen in Fig. 1a. A comparison of Fig. 6a to Fig. 1a demonstrates that the theoretical shallow-water ER wave structure and the structure of the dry ER wave are in good agreement, as the region of maximum ascent lies a quarter wavelength to the east of the low-level cyclonic gyre in both cases. It should be noted, however, that the simulated fields are

TABLE 2. Summary of the WRF simulations used in this study. ADV refers to the horizontal momentum advection terms and A controls the initial ER wave amplitude.

Expt	Dry	Diabatic effects	ADV	A
ER-1	No	Yes	Yes	0.09
ER-2	No	Yes	Yes	0.16
ER-3	No	Yes	Yes	0.23
ER-D-2	Yes	No	Yes	0.16
ER-3-NOADV	No	Yes	No	0.23

² It should be noted that the atmosphere is slightly more stable in the dry simulation (ER-D-2) than in ER-1–ER-3 given that all simulations were initialized with the same base-state temperature lapse rate. To verify that the slight increase in stability was insignificant in the dry simulations, a dry test simulation was run with a slightly less stable lapse rate (results not shown), and results were nearly identical to the ER-D-2 simulation.

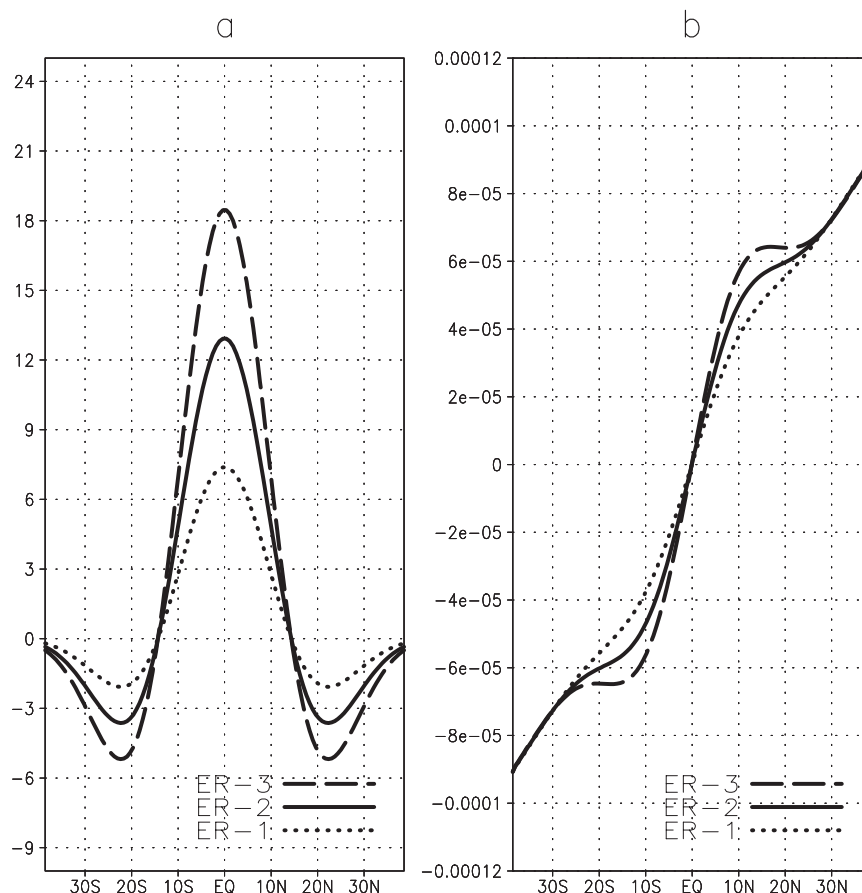


FIG. 4. The initial (a) 850-mb meridional profile of zonal wind (m s^{-1}) for ER-3 (dashed), ER-2 (solid), and ER-1 (dotted) and (b) 850-mb meridional profile of absolute vorticity (s^{-1}). The meridional profiles are centered on the longitude at which the 850-mb relative vorticity is a maximum.

not symmetric about the equator. While the ER wave was initially symmetric, rounding errors and the amplification of these errors owing to nonlinearities led to the development of asymmetries about the equator. When ER-D-2 was run with the momentum advection terms turned off (i.e., limiting the nonlinearities, the simulated ER wave was closer to being symmetric about the equator; results not shown). Additionally, the lack of TC genesis in ER-D-2 demonstrates the significance of diabatic processes with regards to TC genesis.

b. No genesis—Convectively coupled ER waves ER-1 and ER-2

In both the ER-1 and ER-2 simulations, the domain equilibrates over the first 10 days of model integration, as indicated by the changes in the vertical profile of the domain-averaged temperature perturbation (Figs. 7a,b). Between $t = 10$ days and $t = 30$ days, the domain-averaged temperature perturbation remains relatively constant in both simulations, which suggests a quasi-balance

between the surface fluxes, radiation, moist processes, and friction. It should be noted that over the 30-day simulation, the majority of the cooling occurs above 500 mb with a maximum cooling of 4 K near 300 mb. Thus, while there is some drift in the vertical profile of temperature, this result indicates that a quasi-radiative-convective equilibrium has been achieved with the specified radiation scheme.

The zonal wavenumber 10 structure remains intact throughout the entire simulations of ER-1 and ER-2, with the ER wave maintaining a nearly constant phase speed of -2.7 m s^{-1} in both simulations (Figs. 5b,c). The simulated phase speed is about 1 m s^{-1} slower in the westward direction than what was observed in ER-D-2. Additionally, the baroclinic structure in the vertical is maintained throughout the course of the 30-day simulation. Although the magnitude of the meridional wind in ER-1 increases after about $t = 15$ days, the amplitude of the ER wave in ER-2 remains larger than the amplitude of the ER wave in ER-1 from $t = 15$ days to $t = 30$ days.

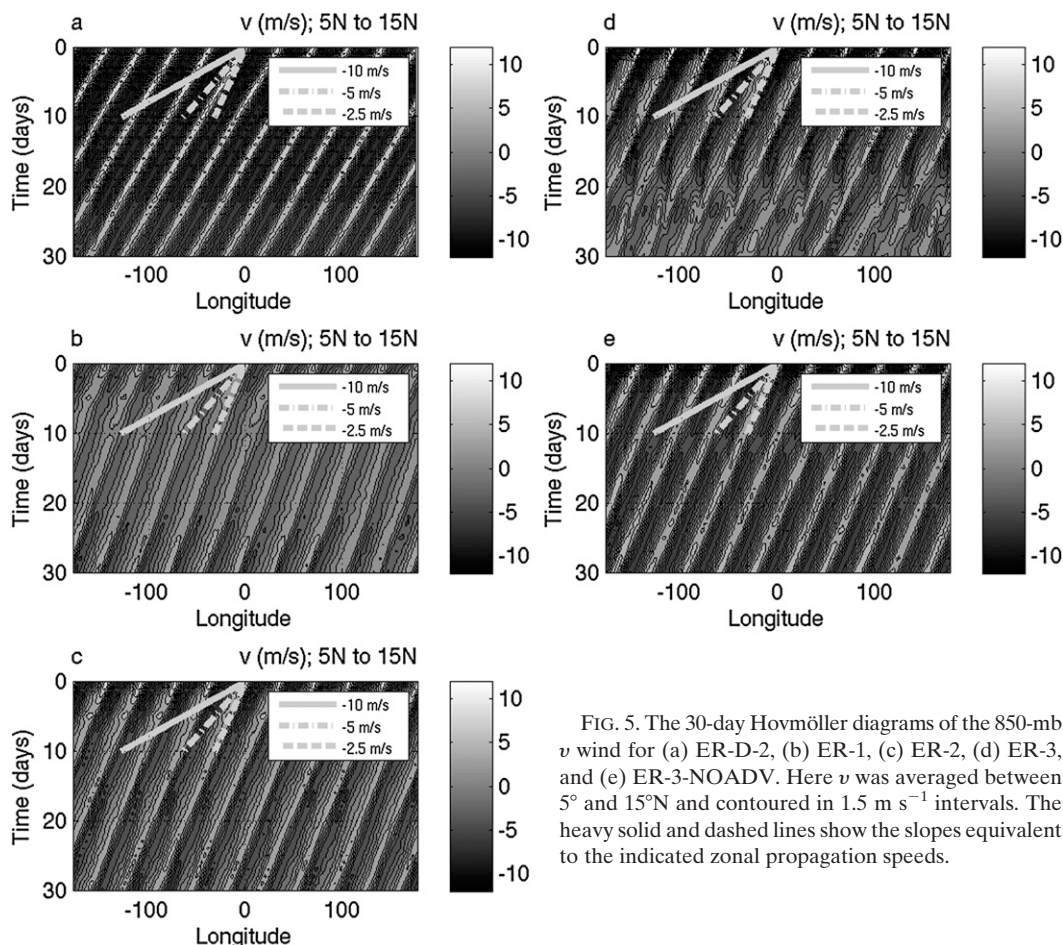


FIG. 5. The 30-day Hovmöller diagrams of the 850-mb v wind for (a) ER-D-2, (b) ER-1, (c) ER-2, (d) ER-3, and (e) ER-3-NOADV. Here v was averaged between 5° and 15°N and contoured in 1.5 m s^{-1} intervals. The heavy solid and dashed lines show the slopes equivalent to the indicated zonal propagation speeds.

The gradual increase in the intensity of the ER wave in ER-1 suggests that the convection is feeding back into the structure of the ER wave.

Figure 8 summarizes the structure of the ER wave from the ER-2 simulation at $t = 30$ days. The ER wave in ER-2 is qualitatively similar to that of ER-1 (figure not shown). The low-level cyclonic gyre is associated with a sea level pressure minimum, and the low-level anticyclonic gyre features a sea level pressure maximum (Fig. 8a). The difference in sea level pressure between the two gyres is only on the order of a few millibars. The weak surface pressure gradient is representative of sea level pressure fluctuations within the tropics often observed with tropical wave activity. As seen in Fig. 8b, the largest low-level (850 mb) relative humidity values are found within the cyclonic portion of the ER wave. This region is associated with a broad region of relative humidity greater than 80%. Since areas of anomalously high low- and midlevel RH are preferred regions for genesis (e.g., Gray 1968), the cyclonic gyre of the ER wave represents a favorable location for genesis relative to the anticyclonic gyre. Finally, the maximum 850-mb

vertical velocities are located to the east of the cyclonic circulation of the ER wave (Fig. 8c). Since vertical velocity is a proxy for convection, most of the convective activity lies in the eastern half of the low-level cyclonic gyre. The location of maximum convective activity coincides within the region of maximum low-level convergence and large-scale ascent, as expected.

Each wavelength of the $t = 30$ -day ER wave from both ER-1 and ER-2 was broken down into 4 quadrants and composited over all 10 wavelengths, as seen in Fig. 9. Quadrants I and II in both simulations featured cyclonic vorticity, while quadrants III and IV were associated with anticyclonic vorticity anomalies about a factor of 3 larger in absolute magnitude (Fig. 9). The western side of the cyclonic gyre and eastern side of the anticyclonic gyre (I and IV) were associated with low-level divergence. The low-level divergence in the western portion of the cyclonic gyre was comparable to that in the eastern portion of the anticyclonic gyre in both ER-1 and ER-2. The eastern half of the cyclonic gyre and western half of the anticyclonic gyre were associated with low-level convergence and mean, deep-level ascent. In this case, the 850-mb

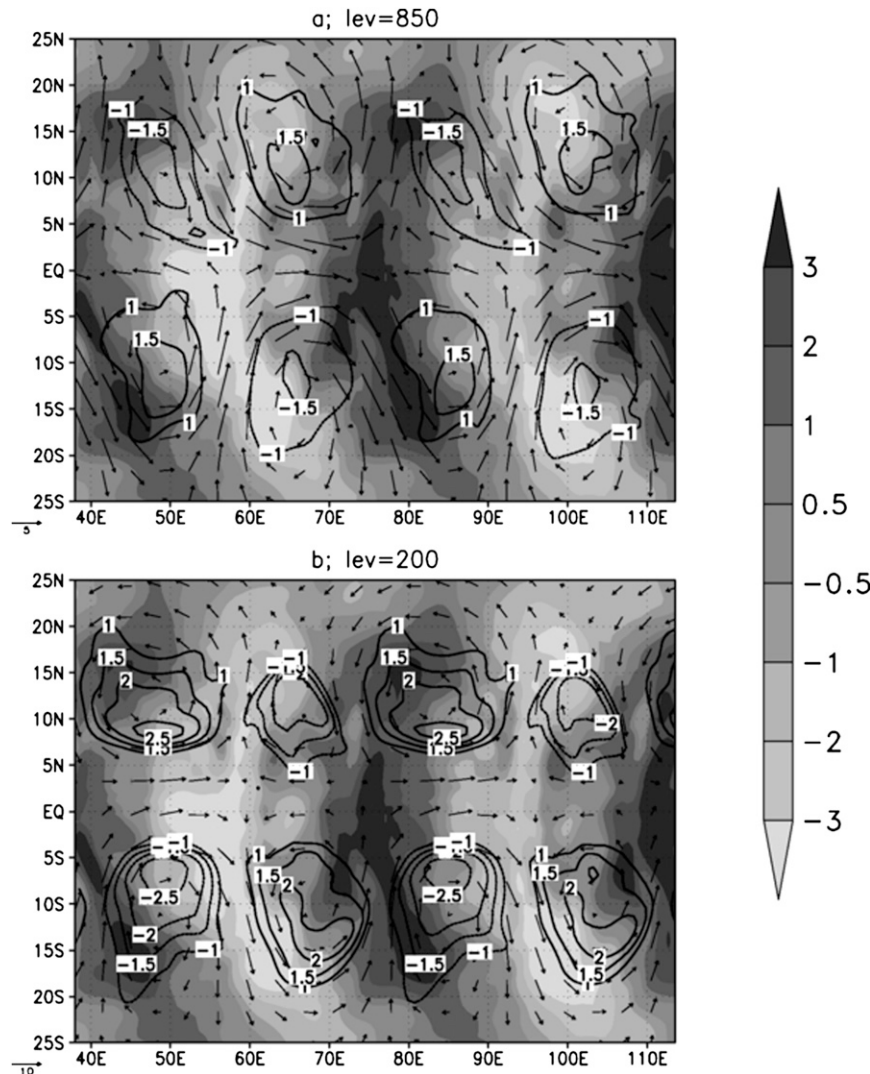


FIG. 6. The $t = 30$ -day ER-D-2 500-mb vertical velocity (10^{-3} m s^{-1} ; shaded) with (a) the 850-mb wind vectors (m s^{-1}) and 850-mb relative vorticity (10^{-5} s^{-1} ; contoured) and (b) the 200-mb wind vectors (m s^{-1}) and 200-mb relative vorticity (10^{-5} s^{-1} ; contoured).

convergence was larger in the cyclonic gyre than in the anticyclonic gyre. The average vertical velocity values reflect the low-level convergence values, as the quadrant-averaged vertical velocity in II was $4.0 \times 10^{-3} \text{ m s}^{-1}$ in ER-2 ($2.9 \times 10^{-3} \text{ m s}^{-1}$ in ER-1), while the quadrant-averaged vertical velocity in III was $0.2 \times 10^{-3} \text{ m s}^{-1}$ ($0.2 \times 10^{-3} \text{ m s}^{-1}$ in ER-1). The magnitude of the 200–850-mb vertical shear was less than 10 m s^{-1} in all four quadrants of the ER wave. Since environmental vertical shear values greater than 10 m s^{-1} are often associated with the weakening of TCs (e.g., Zehr 1992), it was concluded that vertical shear effects were insignificant when compared to vorticity–divergence effects. This result is supported by the findings of Bessafi and Wheeler (2006) and Molinari et al. (2007) as well. Based on the

averaged values of vorticity, divergence, and vertical shear, it is hypothesized that quadrant II features the most favorable conditions for TC genesis within an ER wave since it is within this region in both simulations that anomalous cyclonic relative vorticity, low-level convergence, and weak vertical shear are found. While conditions within certain regions of the ER wave are favorable for TC genesis, it should be noted that genesis is not observed to occur throughout the entire ER-1 or ER-2 simulations.

To examine the sensitivity of the results to horizontal resolution, simulation ER-2 was rerun with a horizontal grid spacing of 27 km (results not shown). The results from this simulation are similar to the ER-2 81-km simulation. Namely, the ER wave phase speed and structure

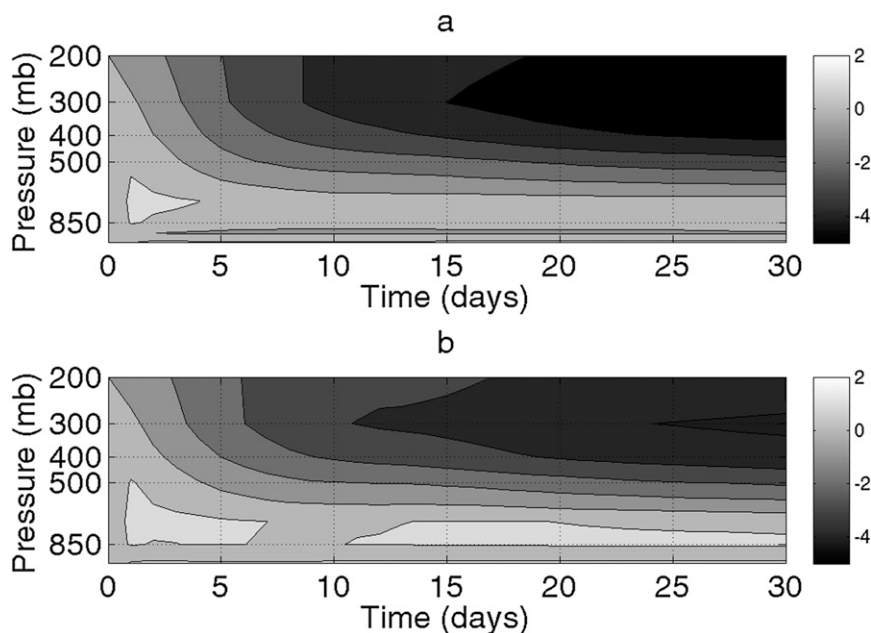


FIG. 7. Vertical profile of the temporal evolution of the domain-averaged temperature perturbation from $t = 0$ for (a) ER-1 and (b) ER-2. Pressure is plotted on a logarithmic scale. The contour interval is 1 K.

were similar, and TC genesis did not occur in either simulation. It is argued that the specified horizontal resolution (81 km) is capable of resolving TC genesis within its global climate model framework (e.g., Stowasser et al. 2007), and is comparable to the grid spacing used in the outermost domain of many limited-area models employed to study various aspects of TC genesis (e.g., Davis and Bosart 2001).

c. Genesis—Convectively coupled ER waves in ER-3 and ER-3-NOADV

The ER-3 850-mb meridional wind Hovmöller diagram (Fig. 5d) is qualitatively similar to both the ER-1 and ER-2 Hovmöller diagrams up until about $t = 18$ days. Past this time, the westward-propagating signal apparent in the ER-3 Hovmöller breaks down owing to the formation of tropical cyclones (Figs. 10a–d). At $t = 20$ days (Fig. 10a), a weak circulation signature is evident in the sea level pressure field. Between $t = 20$ days and $t = 26$ days, the cyclonic circulation intensifies such that by $t = 26$ days, the most intense tropical cyclone has a minimum sea level pressure near 985 mb. In the ER-3-NOADV simulation, however, no tropical cyclogenesis events are observed, and a well-defined, westward-propagating ER wave is evident in the Hovmöller diagram throughout the 30-day simulation (Fig. 5d).

Over the first 11 days of ER-3, the structure of the convectively coupled ER wave remains intact, as exhibited by the 850-mb wind vectors in Fig. 11a. A comparison

of the ER wave from ER-3 to the ER wave from ER-3-NOADV at this time reveals that their structures are qualitatively similar. Three days later at $t = 14$ days, however, the structure in ER-3 begins to exhibit some notable differences from the ER wave in ER-3-NOADV. As denoted in Fig. 11b, an inverted trough oriented in a southwest–northeast direction beginning near the center of the cyclonic gyre of the ER wave is apparent in the 850-mb wind field. In ER-3-NOADV, however, no such deformation of the 850-mb wind field is denoted at this time and location (Fig. 11e). By $t = 17$ days, a closed 850-mb cyclonic circulation with a horizontal scale comparable to that of a TC is located poleward and eastward of the center of the ER wave cyclonic gyre in ER-3. The circulation is centered near 20°N and features a horizontal scale approximately half that of the cyclonic gyre of the ER wave, as seen in Fig. 11c.

The relationship between the horizontal scales of the TC-scale disturbance and the ER wave suggests that a wave self-interaction played a role in the formation of this smaller-scale circulation, as exemplified by the following argument. Suppose that the 850-mb meridional wind is approximated by $A(y) \sin(kx)$ and the zonal wind by $B(y) \cos(kx)$. Since $\partial u / \partial y$ can be written as $\partial B(y) / \partial y [\cos(kx)]$, the product of the meridional wind and $\partial u / \partial y$, that is, the meridional advection of the zonal wind, results in a $\sin(2kx)$ term whose zonal wavenumber is double that of the initial zonal wavenumber. The scale of the resulting cyclonic circulation from ER-3 is an

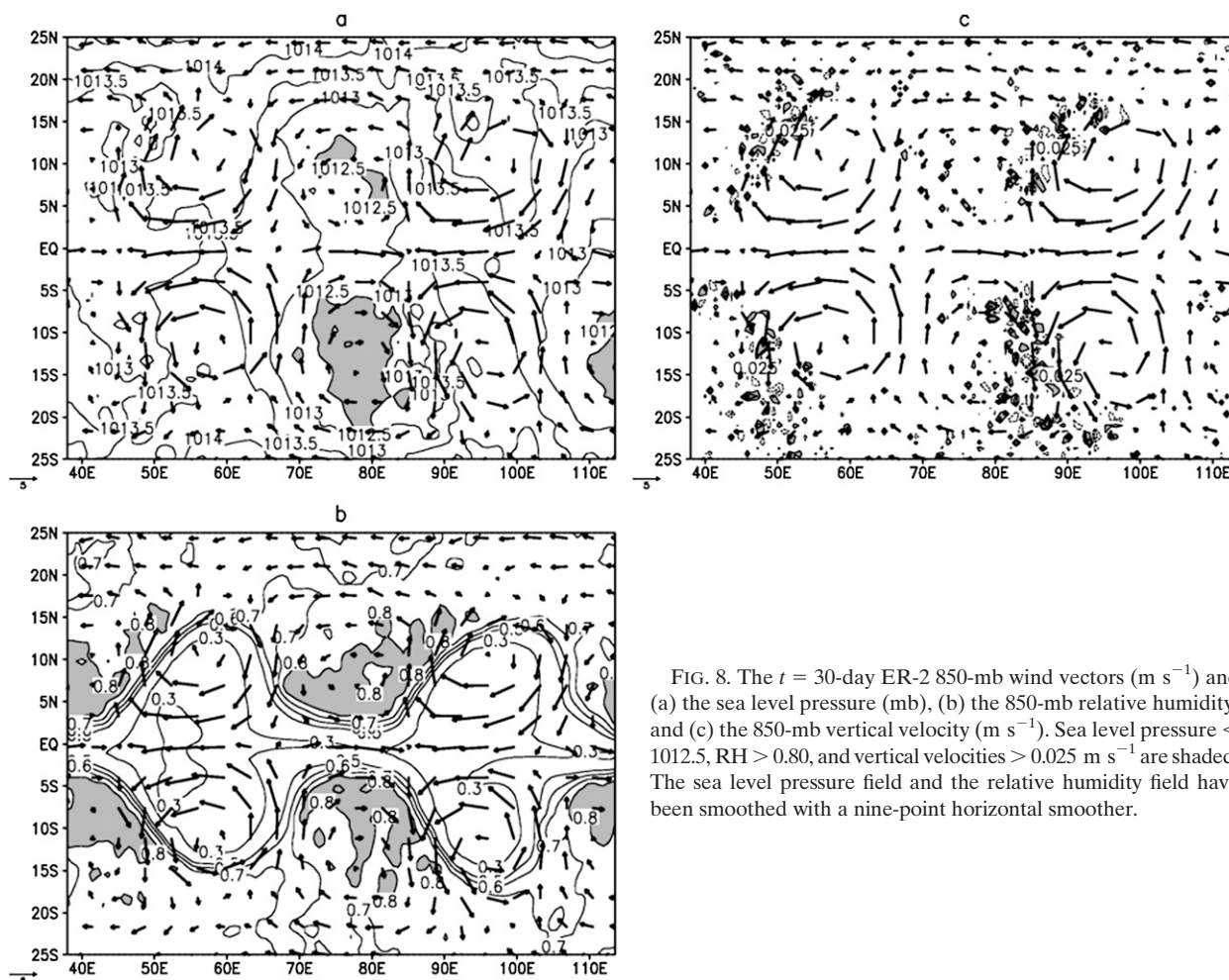


FIG. 8. The $t = 30$ -day ER-2 850-mb wind vectors (m s^{-1}) and (a) the sea level pressure (mb), (b) the 850-mb relative humidity, and (c) the 850-mb vertical velocity (m s^{-1}). Sea level pressure < 1012.5, $\text{RH} > 0.80$, and vertical velocities $> 0.025 \text{ m s}^{-1}$ are shaded. The sea level pressure field and the relative humidity field have been smoothed with a nine-point horizontal smoother.

approximate planetary zonal wavenumber 20, or double the wavenumber of the initial cyclonic circulation of the ER wave. We contend that the nonlinear horizontal momentum advection terms are significant provided that the ER wave is of sufficient amplitude. When the ER-3 simulation is rerun with the horizontal momentum advection terms turned off, no such smaller-scale cyclonic circulation forms as seen in Figs. 11d–f. This result supports our contention that the nonlinear horizontal advection terms play a significant role in tropical cyclogenesis within a sufficiently intense ER wave.

4. Discussion and future work

This study analyzed the structures of simulated ER waves under fixed radiational cooling in a background environment that has no initial mean flow (e.g., no monsoon trough), and examined how these waves might trigger TC genesis. Both the horizontal structure and the baroclinic vertical structure of the ER wave is maintained over the course of the dry simulation (ER-D-2).

Additionally, the large-scale vertical velocity in the dry simulation is a maximum in the eastern half of the low-level cyclonic gyre and the western half of the low-level anticyclonic gyre of the ER wave, and such a result agrees with the large-scale structure predicted from linear theory. For the simulations with moisture, the simulated convectively coupled ER waves are good representations of convectively coupled ER waves found in nature. The -2.7 m s^{-1} phase speed and structure of the simulated ER waves supports the findings of Wheeler and Kiladis (1999), Molinari et al. (2007), and others. One of the main points made in Wheeler and Kiladis (1999) is that the equivalent depths of various convectively coupled waves were observed to be in the range of 12–50 m. The -2.7 m s^{-1} phase speed suggests an equivalent depth toward the low end, but within, this equivalent depth range. The 1 m s^{-1} decrease in the magnitude of the phase speed of the ER wave in ER-1–ER-3 relative to the dry ER wave supports the Wheeler and Kiladis (1999) observation that convective-coupling decreases

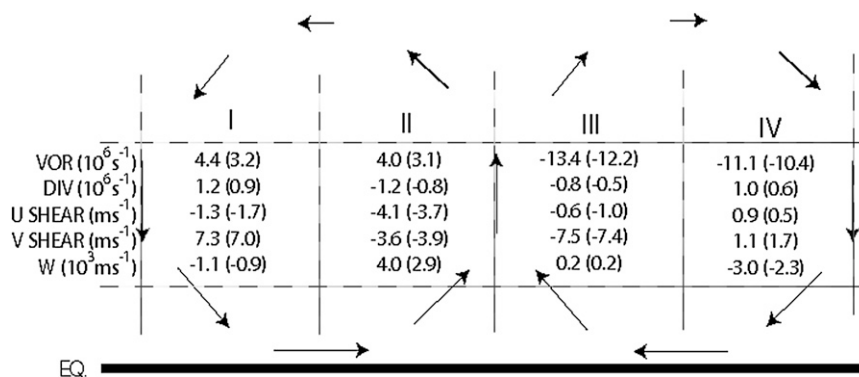


FIG. 9. Four quadrant summary of the ER-2 (ER-1 values in parentheses) $t = 30 \text{ day } 10^{-6} \times 850\text{-mb}$ relative vorticity, $10^{-6} \times 850\text{-mb}$ divergence, 200–850-mb zonal shear, 200–850-mb meridional shear, and $10^{-3} \times 850\text{-mb}$ vertical velocity averaged between 3° and 12°N , and composited over all 10 wavelengths. The quadrant boundaries in the zonal direction were determined by the 850-mb meridional wind local maxima/minima and sign changes at a latitude of 7.5°N .

the propagation speed of tropical waves. The propagation speed of the convectively-coupled ER wave in ER-1 and ER-2 is also similar to the observed -1.9 m s^{-1} propagation speed for a zonal wavenumber 11 ER wave from Molinari et al. (2007).

In both ER-1 and ER-2, the maximum low-level cyclonic vorticity anomalies were on the order of $2.0 \times 10^{-5} \text{ s}^{-1}$ and low-level convergence anomalies were as large as $-1 \times 10^{-5} \text{ s}^{-1}$. Additionally, the magnitude of the vertical shear anomalies was less than 10 m s^{-1} in all four quadrants of the ER waves. The eastern half of the cyclonic gyre of the ER wave contained most of the convection. This location is the preferred region for convection since moist convection is heavily modulated by circulations that cause dynamically forced regions of

vertical motion (e.g., Frank and Ritchie 1999), and such forcing was observed in the eastern half of the cyclonic gyre in the dry simulations. The low-level vorticity, low-level convergence, and weak easterly shear combined with the region of anomalous convection result in conditions most favorable for genesis in the eastern half of the cyclonic gyre of the ER wave.

The TC genesis is only observed to occur for the largest-amplitude convectively coupled ER wave. We argue that genesis in this simulation is due to the large magnitudes of the nonlinear horizontal momentum advection terms, or the so-called wave self-interactions. In the smaller initial-amplitude ER wave simulations (ER-1 and ER-2), it is hypothesized that the smaller-scale cyclonic circulations, with horizontal wavelengths half

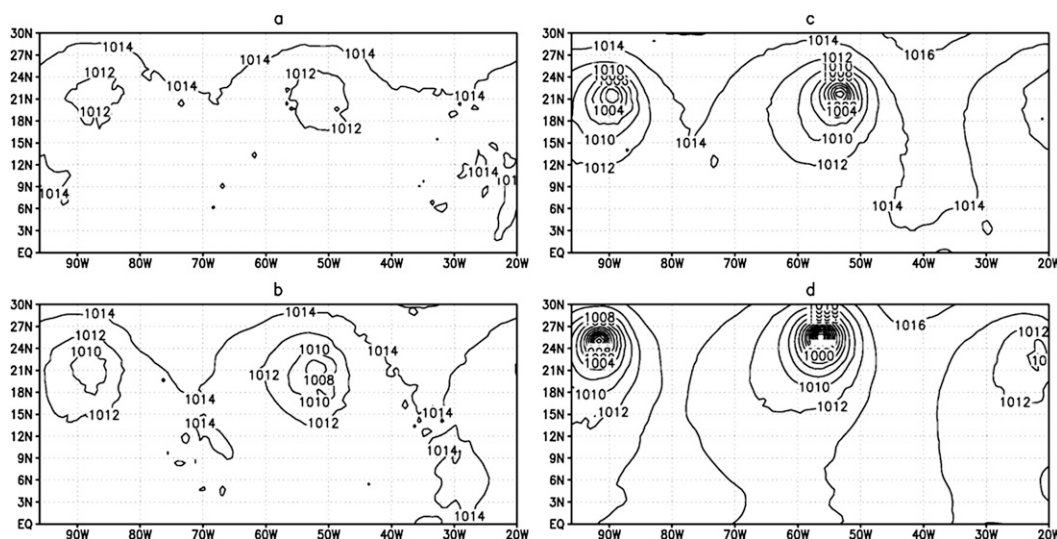


FIG. 10. The ER-3 sea level pressure field (mb) over two arbitrary wavelengths at (a) $t = 20$, (b) 22, (c) 24, and (d) 26 days.

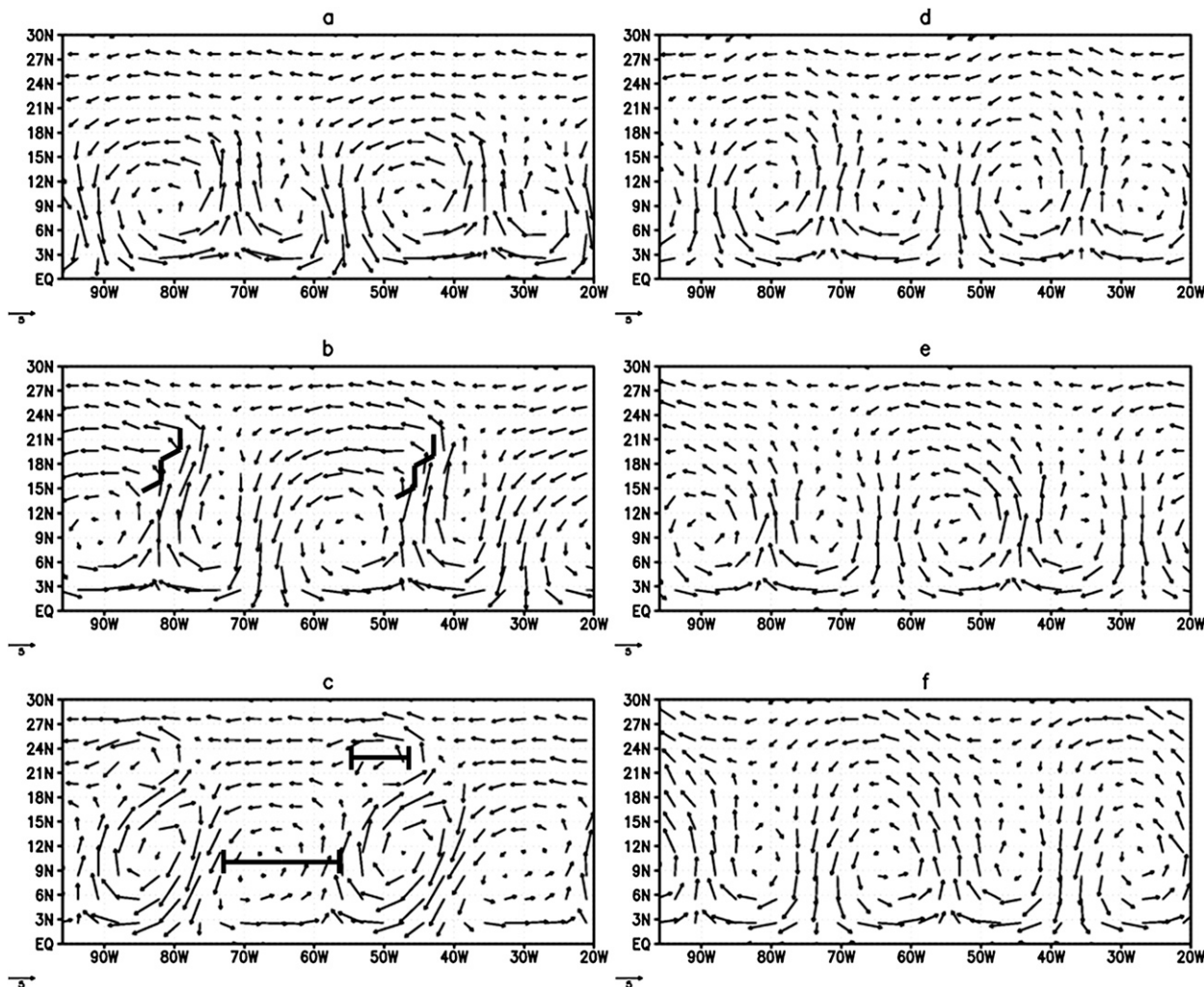


FIG. 11. (a)–(c) The ER-3 850-mb wind vectors (m s^{-1}) and (d)–(f) ER-3-NOADV 850-mb wind vectors (m s^{-1}) at (a),(d) $t = 11$; (b),(e) $t = 14$; and (c),(f) $t = 17$ days. The jagged lines in (b) denote the locations of the inverted troughs. The long solid line in (c) indicates the zonal scale of the cyclonic gyre of the ER wave and the solid short line indicates the zonal scale of the smaller-scale cyclonic disturbance.

that of the cyclonic gyre of the ER wave, never form because the magnitude of the nonlinear horizontal advection terms remain sufficiently small (i.e., the amplitude of the ER wave remains below some threshold amplitude).

While we are not dismissing this genesis mechanism within an ER wave, we contend that this is not the typical pathway to genesis within an ER wave. First, it should be noted that the initial amplitude of the ER wave from the ER-3 simulation was close to being barotropically unstable. This initial relative vorticity maximum near $3 \times 10^{-5} \text{ s}^{-1}$ may be unrealistically large, as the maximum anomalous relative vorticity values derived from observational ER wave-filtered studies (e.g., Frank and Roundy 2006; Molinari et al. 2007; Bessafi and Wheeler 2006) are all smaller in magnitude. Second, the location of genesis relative to the ER wave lies well

poleward and eastward of the center of the ER wave cyclonic gyre. Recent ER wave composites of Frank and Roundy (2006) and Molinari et al. (2007) relative to a mean genesis location, however, demonstrated that genesis occurred within the eastern half of the cyclonic gyre of the ER wave in both studies. The approximate mean genesis locations from these studies lie about a quarter wavelength to the west of and equatorward of the genesis location observed in the ER-3 simulation.

We hypothesize that a much more common mechanism for genesis within an ER wave is due to the interaction of a sufficiently intense convectively coupled ER wave with a favorable background environment (e.g., a monsoon trough). This interactive genesis mechanism is examined in detail in Part II of this series of papers in which a convectively coupled ER wave that does not

result in genesis (ER-2) is initialized in different idealized background flow configurations.

Owing to the uniqueness of the methodology employed herein, there remains a plethora of unanswered questions that are not addressed in this paper or in the complementary Part II study. For example, only a planetary zonal wavenumber 10 ER wave was considered. We plan to conduct a suite of sensitivity studies in which certain parameters (e.g., wavenumber and SST) are varied and examine how the phase speed as well as the convectively coupled structure of the ER wave changes. Additionally, the sensitivity of ER wave-induced TC genesis to the radiation scheme will be examined in more detail. It is possible that a more realistic radiation scheme may promote convective self-aggregation, which in turn, may make TC genesis within an ER wave more likely than presented here.

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REFERENCES

- Bessafi, M., and M. C. Wheeler, 2006: Modulation of South Indian Ocean tropical cyclones by the Madden–Julian oscillation and convectively coupled equatorial waves. *Mon. Wea. Rev.*, **134**, 638–656.
- Bretherton, C. S., P. N. Blossey, and N. Khairoutdinov, 2005: An energy-balance analysis of deep convective self-aggregation above uniform SST. *J. Atmos. Sci.*, **62**, 4273–4292.
- Chen, S.-H., and J. Dudhia, 2000: Annual report. WRF Physics, Air Force Weather Agency, 38 pp. [Available online at box.mmm.ucar.edu/wrf/users/docs/wrf-doc-physics.pdf.]
- Davis, C., and L. F. Bosart, 2001: Numerical simulations of the genesis of Hurricane Diana (1984). Part I: Control simulation. *Mon. Wea. Rev.*, **129**, 1859–1881.
- Delaten, K., and J.-I. Yano, 2009: Is asymptotic nondivergence of the large-scale tropical atmosphere consistent with equatorial wave theories? *Tellus*, **61A**, 491–497.
- Frank, W. M., and E. A. Ritchie, 1999: Effects of environmental flow upon tropical cyclone structure. *Mon. Wea. Rev.*, **127**, 2044–2061.
- , and P. E. Roundy, 2006: The role of tropical waves in tropical cyclogenesis. *Mon. Wea. Rev.*, **134**, 2397–2417.
- Gall, J. S., and W. M. Frank, 2010: The role of equatorial Rossby waves in tropical cyclogenesis. Part II: Idealized simulations in a monsoon trough environment. *Mon. Wea. Rev.*, **138**, 1383–1398.
- Gray, W. M., 1968: Global view of the origin of tropical disturbances and storms. *Mon. Wea. Rev.*, **96**, 669–700.
- Holton, J. R., 2004: *Introduction to Dynamic Meteorology*. 4th ed. International Geophysics Series, Vol. 88, Academic Press, 535 pp.
- Hong, S. Y., and H. L. Pan, 1996: Nonlocal boundary layer vertical diffusion in a medium-range forecast model. *Mon. Wea. Rev.*, **124**, 2322–2339.
- , J. Dudhia, and S. H. Chen, 2004: A revised approach to ice microphysical processes for the bulk parameterization of clouds and precipitation. *Mon. Wea. Rev.*, **132**, 103–120.
- Jordan, C. L., 1958: Mean soundings for the West Indies area. *J. Meteor.*, **15**, 91–97.
- Kain, J. S., and J. M. Fritsch, 1990: A one-dimensional entraining/detraining plume model and its application in convective parameterization. *J. Atmos. Sci.*, **47**, 2784–2802.
- , and —, 1993: Convective parameterization for mesoscale models: The Kain–Fritsch scheme. *The Representation of Cumulus Convection in Numerical Models*, Meteor. Monogr., No. 24, Amer. Meteor. Soc., 165–170.
- Kiladis, G. N., and M. C. Wheeler, 1995: Horizontal and vertical structure of observed tropospheric equatorial Rossby waves. *J. Geophys. Res.*, **100**, 22 981–22 998.
- Lin, Y. L., R. D. Farley, and H. D. Orville, 1983: Bulk parameterization of the snow field in a cloud model. *J. Climate Appl. Meteor.*, **22**, 1065–1092.
- Madden, R. A., and P. R. Julian, 1994: Observations of the 40–50-day tropical oscillation. *Mon. Wea. Rev.*, **122**, 814–837.
- Matsuno, T., 1966: Quasi-geostrophic motions in the equatorial area. *J. Meteor. Soc. Japan*, **44**, 25–43.
- Michalakes, J., S. Chen, J. Dudhia, L. Hart, J. Klemp, J. Middlecoff, and W. Skamarock, 2001: Development of a next generation regional Weather Research and Forecast model. *Developments in Teracomputing: Proceedings of the Ninth ECMWF Workshop on the Use of High Performance Computing in Meteorology*, W. Zwielfhofer and N. Kreitz, Eds., Vol. 1, World Scientific, 269–276.
- Molinari, J., K. Lombardo, and D. Vollaro, 2007: Tropical cyclogenesis within an equatorial Rossby wave packet. *J. Atmos. Sci.*, **64**, 1301–1317.
- Noh, Y., W. G. Chun, S. Y. Hong, and S. Raasch, 2004: Improvement of the k-profile model for the planetary boundary layer based on large eddy simulation data. *Bound.-Layer Meteor.*, **107**, 401–427.
- Reasor, P. D., M. T. Montgomery, and L. F. Bosart, 2005: Mesoscale observations of the genesis of Hurricane Dolly. *J. Atmos. Sci.*, **62**, 3151–3171.
- Roundy, P. E., and W. M. Frank, 2004a: A climatology of waves in the equatorial region. *J. Atmos. Sci.*, **61**, 2105–2132.
- Stowasser, M., Y. Wang, and K. Hamilton, 2007: Tropical cyclone changes in the western North Pacific in a global warming scenario. *J. Climate*, **20**, 2378–2396.
- Wheeler, M. C., 2002: Tropical meteorology: Equatorial waves. *Encyclopedia of Atmospheric Sciences*, J. Holton, J. Curry, and J. Pyle, Eds., Academic Press, 2313–2325.
- , and G. N. Kiladis, 1999: Convectively coupled equatorial waves: Analysis of clouds and temperature in the wavenumber-frequency domain. *J. Atmos. Sci.*, **56**, 374–399.
- , —, and P. J. Webster, 2000: Large-scale dynamical fields associated with convectively coupled equatorial waves. *J. Atmos. Sci.*, **57**, 613–640.
- Zehr, R. M., 1992: Tropical cyclogenesis in the western North Pacific. NOAA Tech. Rep. NESDIS 61, 181 pp.
- Zhang, C., 2005: Madden–Julian oscillation. *Rev. Geophys.*, **43**, 1–36.